

PHASE 2 – REQUIREMENT ANALYSIS

2.1 Introduction

Requirement analysis identifies the functional and technical requirements necessary for developing the AI BlogNest platform.

The requirements are divided into functional requirements, non-functional requirements, software requirements, and hardware requirements.

2.2 Functional Requirements

The system should provide the following functions:

User Management

- User registration.
- User login.
- Secure password storage.
- User authentication.
- Role-based authorization.
- User role management.

Blog Management

- Create blog posts.
- View blog posts.
- Update existing blog posts.
- Delete blog posts.
- Manage blog publishing status.
- Organize posts using categories and tags.

AI Features

- Generate blog content using AI.
- Generate FAQ-related assistance.
- Generate content insights.
- Provide AI-based summarization.
- Generate context-aware responses.

Comment Management

- Allow users to add comments.
- Retrieve comments.
- Manage comments.
- Support moderation of inappropriate comments.

Search and Organization

- Search blog content.
- Filter content using categories and tags.
- Organize blog information efficiently.

2.3 Non-Functional Requirements

Security

The system should protect user information through JWT-based authentication, password hashing, input validation, sanitization, and role-based authorization.

Performance

API requests should be processed efficiently and database operations should be optimized through suitable data models.

Scalability

The modular backend architecture should allow additional services and features to be incorporated in the future.

Reliability

The application should provide centralized error handling and maintain consistent API responses.

Maintainability

The application should be organized into separate models, controllers, routes, middleware, and services.

2.4 Software Requirements

- Operating System: Windows 10/11, macOS, or Linux.
- Node.js: JavaScript runtime environment.
- npm: Package management system.
- Express.js: Backend web framework.
- MongoDB: NoSQL database.
- Mongoose: MongoDB object data modeling library.
- Google Gemini AI: Generative AI service.
- Postman / Thunder Client: API testing.
- Visual Studio Code: Development environment.

2.5 Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or equivalent.

- RAM: Minimum 8 GB.
- Storage: Minimum 1 GB of available space.
- Internet connection for AI service and database connectivity.

2.6 User Requirements

The system supports different categories of users:

- Administrator
- Editor
- Author
- Reader

Each role receives access according to its responsibilities.